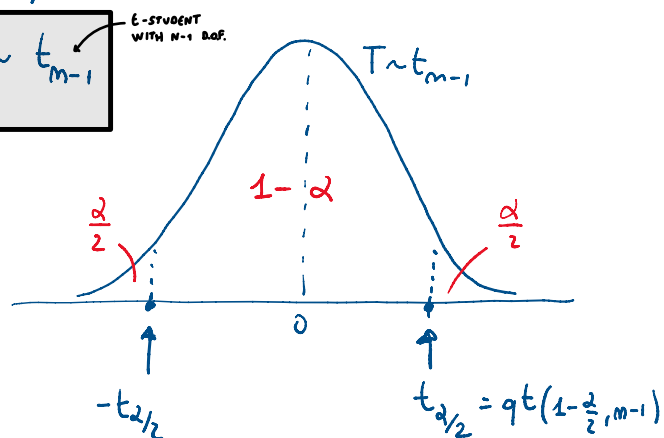


WE ARE STUDYING CONFIDENCE INTERVALS (CI)

WE HAVE DERIVED $CI_{1-\alpha}(\mu)$ WHEN $X \sim N(\mu, \sigma^2)$ AND σ^2 UNKNOWN

THE PIVOT QUANTITY IS $T = \frac{\bar{X} - \mu}{S/\sqrt{m}} \sim t_{m-1}$

(CENTRAL) CIs SPLIT α PROBABILITY
EVENLY IN THE TWO TAILS



$$\dots \Rightarrow CI_{1-\alpha}(\mu) = \left[\bar{X} \pm t_{\alpha/2} \frac{S}{\sqrt{m}} \right]$$

POINT ESTIMATOR

MARGIN OF ERROR

INTERVAL LENGTH = UPPER BOUND - LOWER BOUND = $L_2 - L_1 = \left(\bar{X} + t_{\alpha/2} \frac{S}{\sqrt{m}} \right) - \left(\bar{X} - t_{\alpha/2} \frac{S}{\sqrt{m}} \right)$

$$= 2 t_{\alpha/2} \frac{S}{\sqrt{m}}$$

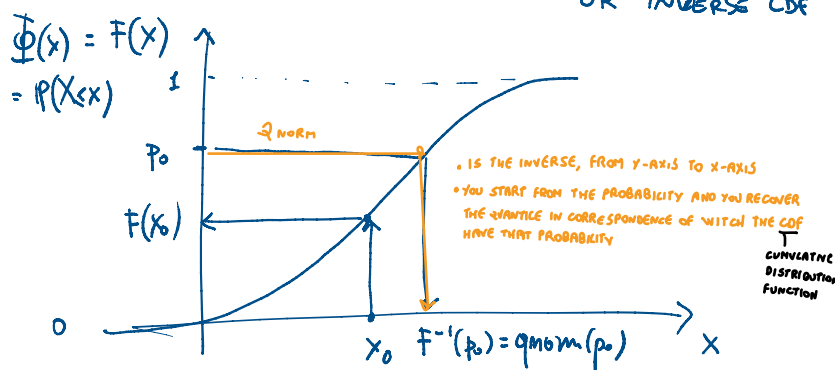
WE CAN COMPARE THIS WITH LENGTH IN PREVIOUS EXAMPLE (σ^2 known)

$$\left(2 z_{\alpha/2} \frac{\sigma}{\sqrt{m}} \right) \quad \text{PREVIOUS EXAMPLE}$$

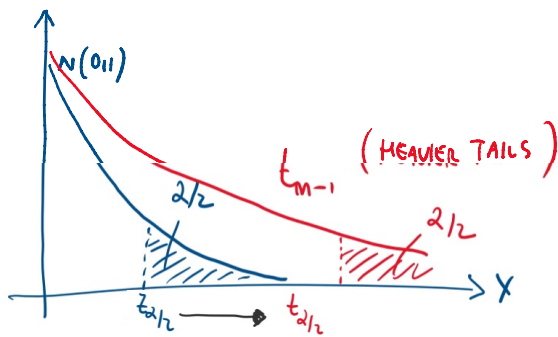
$\Rightarrow$ NOW THE STATISTICAL PROBLEM IS NONE DIFFICULT SINCE I HAVE LESS INFO

$\Rightarrow$ I EXPECT A WIDER CI AT A GIVEN LEVEL OF CONF. (BECAUSE OF t-STUDENT QUANTILE)

PARENTHESIS ON $q_{\text{norm}} = F^{-1}(p)$ QUANTILE FUNCTION OR INVERSE CDF



$\Rightarrow$ ZOOM ON THE RIGHT-TAIL:



$t_{\alpha/2} > z_{\alpha/2} \Rightarrow$ I HAVE NOW A HIGHER MARGIN OF ERROR

$\Rightarrow$ A WIDER CI (IN EXPECTATION)

A SECOND DIFFERENCE WRT THE CASE σ^2 KNOWN : $A = L_2 - L_1 = 2 t_{\alpha/2} \frac{S}{\sqrt{n}}$

$\Rightarrow$ WHEN σ UNKNOWN, CI LENGTH IS RANDOM

NOW DEPENDS ON $\underline{X}$

$\Rightarrow$ I CAN'T BE SURE THAT A MINIMUM n WILL DETERMINE A MARGIN OF ERROR BELOW A DESIRED THRESHOLD

(BEFORE $A = 2 t_{\alpha/2} \frac{\sigma}{\sqrt{n}}$)

LAST TIME: WE WANTED $A < 10 \Leftrightarrow 2 t_{\alpha/2} \frac{\sigma}{\sqrt{n}} < 10$

$$\Rightarrow \sqrt{n} > 2 t_{\alpha/2} \frac{\sigma}{10} \Rightarrow n > \left(2 t_{\alpha/2} \frac{\sigma}{10} \right)^2$$

NOW: IF I WANT $A < 10 \Rightarrow 2 t_{\alpha/2} \frac{S}{\sqrt{n}} < 10$

$$\Rightarrow \sqrt{n} > 2 t_{\alpha/2} \frac{S}{10} \Rightarrow n > \left(2 t_{\alpha/2} \frac{S}{10} \right)^2$$

UNKNOWN IN ADVANCE

EX:

WE WANT TO STUDY THE EFFECT OF MERGERS ON REVENUES

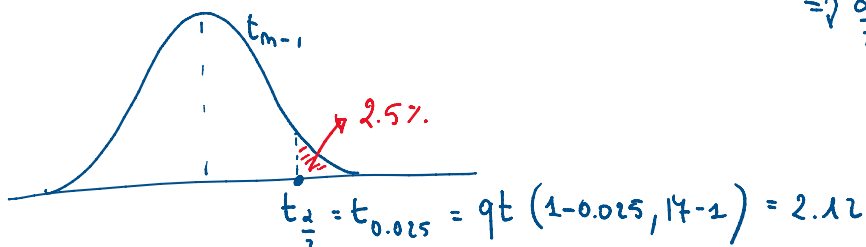
X : DIFFERENCE IN REVENUE POST - PRE MERGER

$n=14$, $\bar{X}=0.105$, $S=0.44$. FIND $CI_{95\%}(\mu)$

WE ASSUME $X \sim \text{NORMAL}$

$$CI_{1-\alpha}(\mu) = \left[\bar{X} \pm t_{\alpha/2} \frac{S}{\sqrt{n}} \right] \quad . \quad 1-\alpha = 0.95 \Rightarrow \alpha = 0.05$$

$$\Rightarrow \frac{\alpha}{2} = 0.025$$



$$\Rightarrow CI_{95\%}(\mu) = \left[0.105 \pm 2.12 \cdot \frac{0.44}{\sqrt{14}} \right] = [-0.121, +0.331]$$

WE SEE NO STATISTICAL EVIDENCE OF BETTER OR WORSE REVENUES FOLLOWING MERGERS, SINCE

BOTH NEGATIVE AND POSITIVE VALUES OF X ARE

"REASONABLE" (THEY BOTH BELONG TO THE CI)

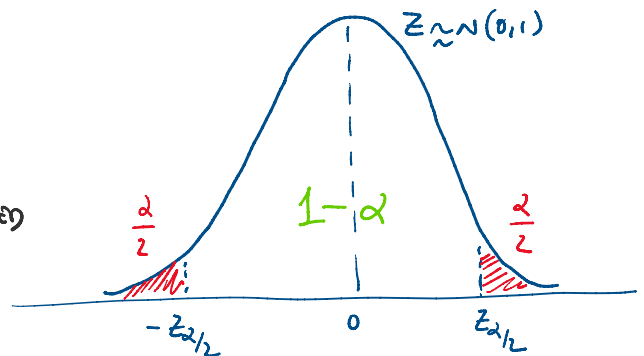
$$\text{SUMMARIZING: } X \sim N(\mu, \sigma^2) \begin{cases} \sigma^2 \text{ KNOWN} \rightarrow C_{1-\alpha}(\mu) = [\bar{X} \mp z_{\alpha/2} \frac{\sigma}{\sqrt{n}}] \\ \sigma^2 \text{ UNKNOWN} \rightarrow C_{1-\alpha}(\mu) = [\bar{X} \mp t_{\alpha/2} \frac{s}{\sqrt{n}}] \end{cases}$$

✓ POINT ESTIMATION

NOW: REMOVE ASSUMPTION OF NORMALITY, BUT "LARGE SAMPLE"
 ($n \geq 30$ WITH σ^2 KNOWN, $n \geq 50$ WITH σ^2 UNKNOWN)

$C_{1-\alpha}(\mu)$, $X \sim \text{UNKNOWN}$, n LARGE

PQ IS $Z = \frac{\bar{X} - \mu}{s/\sqrt{n}} \approx N(0, 1)$
↓ PIVOTAL QUANTITY



$P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) \approx 1-\alpha$
APPROXIMATELY DISTRIBUTED AS NORMAL
APPROXIMATELY EQUAL BECAUSE APPROXIMATELY NORMAL

$\Rightarrow P(-z_{\alpha/2} \leq \frac{\bar{X} - \mu}{s/\sqrt{n}} \leq z_{\alpha/2}) \approx 1-\alpha$

$\Rightarrow P(-z_{\alpha/2} \frac{s}{\sqrt{n}} \leq \bar{X} - \mu \leq z_{\alpha/2} \frac{s}{\sqrt{n}}) \approx 1-\alpha$

$\Rightarrow P(\underbrace{\bar{X} - z_{\alpha/2} \frac{s}{\sqrt{n}}}_{L_1 \text{ LOWER BOUND}} \leq \underbrace{\mu}_{\theta} \leq \underbrace{\bar{X} + z_{\alpha/2} \frac{s}{\sqrt{n}}}_{L_2 \text{ UPPER BOUND}}) \approx 1-\alpha$

$P(L_1 < \theta < L_2) \approx 1-\alpha$

$\Rightarrow C_{1-\alpha}(\mu) \approx [\bar{X} \mp z_{\alpha/2} \frac{s}{\sqrt{n}}]$

"IN THE MIDDLE" OF THE TWO PREVIOUS EXAMPLES

EX:

X: "JOB SECURITY" EVALUATION

$\text{supp}(X) = \{1, 2, \dots, 5\}$

$n = 142$

$\bar{X} = 4.38$, $S = 0.40$. ON AVERAGE ✓ SAMPLE STANDARD DEVIATION

FIND $C_{99\%}(\mu)$

VERY IMPORTANT JOB SECURITY
 NOT IMPORTANT

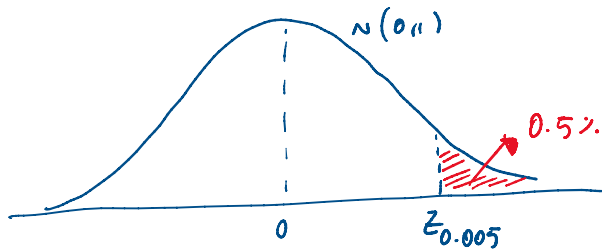
$\bar{X} = 4.38$, $S = 0.40$. FIND $C_{99\%}(\mu)$

$$C_{99\%}(\mu) \approx \left[\bar{X} \pm z_{\alpha/2} \frac{S}{\sqrt{n}} \right]$$

CONF. LEVEL = $1 - \alpha = 0.99$

$\Rightarrow \alpha = 0.01$

$\Rightarrow \frac{\alpha}{2} = 0.005 = 0.5\%$



$z_{\alpha/2} = z_{0.005} = \text{qnorm}(1 - 0.005)$
 $= 2.575$

$\Rightarrow C_{99\%}(\mu) \approx \left[4.38 \pm 2.575 \cdot \frac{0.40}{\sqrt{172}} \right] = [4.24, 4.52]$

I expect TRUE μ TO BE IN HERE WITH 99% CONFIDENCE

BERNOLLI SUCCESS RATE

$$C_{1-\alpha}(p), X \sim \text{Bern}(p), n \text{ LARGE}$$

POINT ESTIMATOR: $\hat{\theta} - \theta$ (TRUE ESTIMATOR)

STANDARD DEVIATION: $\sqrt{\hat{\sigma}_{\hat{\theta}}^2}$

ESTIMATED STANDARD ERROR OF THE POINT ESTIMATOR $\hat{\theta}$

NOW! $\theta = p$ (ESTIMATE)

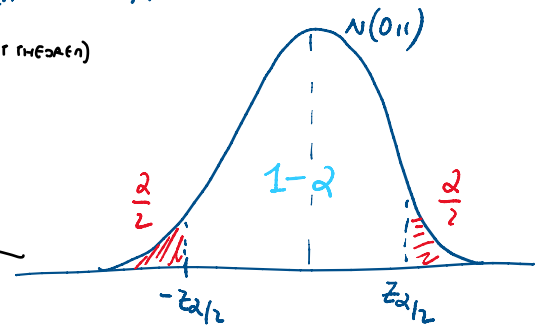
$\hat{\theta} = \bar{X}$

$\sigma_{\hat{\theta}}^2 = V(\bar{X}) = \frac{1}{n} V(X) = \frac{1}{n} p(1-p)$

$\hat{\sigma}_{\hat{\theta}}^2 = \hat{V}(\bar{X}) = \frac{1}{n} \hat{V}(X) = \frac{1}{n} \bar{X}(1-\bar{X})$

APPROXIMATELY TRUE BY CLT (CENTRAL LIMIT THEOREM) IN LARGE SAMPLE

$\Rightarrow Z = \frac{\bar{X} - p}{\sqrt{\frac{1}{n} \bar{X}(1-\bar{X})}} \approx N(0,1)$



$P(-z_{\alpha/2} \leq Z \leq +z_{\alpha/2}) \approx 1 - \alpha$

$\Rightarrow P\left(-z_{\alpha/2} \leq \frac{\bar{X} - p}{\sqrt{\frac{1}{n} \bar{X}(1-\bar{X})}}} \leq +z_{\alpha/2}\right) \approx 1 - \alpha$

$\frac{A}{B} = \frac{p}{c} = \frac{A \cdot c}{B}$ $\frac{A}{B} = \frac{p}{c} = \frac{A \cdot c}{B}$

$\frac{A}{B} = \frac{p}{c} = \frac{A \cdot c}{B}$

... ('KEEP' p IN THE MIDDLE) ...

$\Rightarrow P\left(\underbrace{\bar{X} - z_{\alpha/2} \sqrt{\frac{\bar{X}(1-\bar{X})}{n}}}_{L_1} \leq \underbrace{p}_{\theta} \leq \underbrace{\bar{X} + z_{\alpha/2} \sqrt{\frac{\bar{X}(1-\bar{X})}{n}}}_{L_2}\right) \approx 1 - \alpha$

$$\Rightarrow C_{1-\alpha}(p) \approx \left[\bar{x} \pm z_{\alpha/2} \sqrt{\frac{\bar{x}(1-\bar{x})}{m}} \right]$$

EX: X : YES/NO ANSWER TO "DO YOU ACCEPT GIFTS FROM VENDORS?"

83 ANSWERS ARE "NO" OUT OF A TOT OF 344 = m SAMPLE SIZE

ARBITRARY IN BERNULLI
 $X = \begin{cases} 0, & \text{IF "YES"} \\ 1, & \text{IF "NO"} \end{cases}$ 83 FIND $C_{90\%}(p)$

p : "TRUE POP PROPORTION OF PEOPLE WHO DO NOT ACCEPT GIFTS"

$$\begin{aligned} C_{90\%}(p) &= \left[\bar{x} \pm z_{\alpha/2} \sqrt{\frac{\bar{x}(1-\bar{x})}{m}} \right] \\ &= \left[\frac{83}{344} \pm 1.645 \sqrt{\frac{83/344 \cdot (1-83/344)}{344}} \right] \\ &\quad \underbrace{1 - \frac{0.10}{2}} \\ &= [0.203, 0.249] \end{aligned}$$

IF I COLLECT $N=344$ ANSWERS AND 83 OF THEM ARE NO, THE TRUE PROPORTION OF PEOPLE WITH A CONFIDENCE LEVEL OF 90% THAT DO NOT ACCEPT GIFTS IS BETWEEN 0.203 AND 0.249.

EX: p : "POP PROPORTION OF CLIENTS INTERESTED TO A NEW SOFTWARE"

$X = \begin{cases} 1, & \text{IF INTERESTED} \\ 0, & \text{IF NOT} \end{cases}$

FIND m SAMPLE SIZE SUCH THAT MSE OF $\hat{p} = \bar{X}$ IS SMALLER THAN 0.005 MEAN SQUARE ERROR

$$\text{MSE}_p(\bar{X}) = \underset{\substack{\uparrow \\ \bar{X} \text{ UNBIASED}}}{V(\bar{X})} = \frac{1}{m} V(X) = \frac{p(1-p)}{m}$$

$$\Rightarrow \text{I WANT } m \text{ S.T. } \frac{p(1-p)}{m} < 0.005 \Rightarrow m > \frac{p(1-p)}{0.005}$$

PROBLEM: I DON'T KNOW $p \Rightarrow$ I DON'T KNOW THIS THRESHOLD FOR m

$\Rightarrow$ PROVIDE A "CONSERVATIVE" ANSWER WHICH IS VALID LITTLE LESS PRECISE BUT VALID IN ALL CASES FOR ALL POSSIBLE VALUES OF p

$\Rightarrow$ INSTEAD OF FIXING $\frac{p(1-p)}{n} < 0.005$, 1 FIX

$$\max_p \frac{p(1-p)}{n} < 0.005$$

$$f(p) := \frac{p(1-p)}{n} = \frac{p-p^2}{n} \quad \frac{\partial}{\partial p} f(p) = \frac{1-2p}{n} \stackrel{!}{=} 0 \Rightarrow p^* = 1/2$$

$$\frac{d^2}{dp^2} f(p) = -2/m < 0 \Rightarrow p^* = 1/2 \text{ is that } p \text{ at which } MSE_p(\bar{x}) \text{ is max}$$

$$MSE_p(\bar{x}) \Big|_{p=1/2} = \frac{1/2(1-1/2)}{n} = \frac{1}{4n} \quad (\text{Worst-CASE SCENARIO})$$

$$\Rightarrow \text{Fix } \frac{1}{4m} < 0.005 \Rightarrow m > \frac{1}{4 \cdot 0.005} = 50$$

Now: we collect $m=100$, with $\bar{x}=0.10$

PREK NSE_p($\bar{x}$) and find C(99%) (p)

$$PSE_p(\bar{x}) = p(1-p)/m \quad \widehat{PSE}_p(\bar{x}) = \bar{x}(1-\bar{x})/m = \frac{0.10 \cdot 0.90}{100} = 0.009 < 0.005$$

$$CI_{99\%}(p) = \left[\bar{X} \mp z_{\alpha/2} \sqrt{\frac{\bar{X}(1-\bar{X})}{n}} \right]$$

$$= \left[0.10 \mp 2.576 \left(1 - \frac{d}{2} \right) \sqrt{\frac{0.10 \cdot 0.90}{100}} \right] = [0.0227, 0.1773]$$

$\alpha = 0.01$

$$1 - \frac{\alpha}{2} = 0.995$$

MOVE FROM ONE SINGLE POPULATION TO A COUPLE OF POPULATION

$$C_{1-\alpha}(\mu_1 - \mu_2), \quad X \sim N(\mu_1, \sigma_1^2), \quad Y \sim N(\mu_2, \sigma_2^2), \quad X \perp Y$$

X and Y
ARE
INDEPENDENT

1) σ_1^2 AND σ_2^2 KNOWN . $\underline{X} = (X_1, \dots, X_{m_1})$, $\underline{Y} = (Y_1, Y_2, \dots, Y_{m_2})$

THE TWO SAMPLES X AND Y CAN HAVE THEIR OWN DIFFERENT SIZES

$$\bar{X} \sim N(\mu_1, \sigma_1^2/m_1) \quad \text{AND} \quad \bar{Y} \sim N(\mu_2, \sigma_2^2/m_2)$$

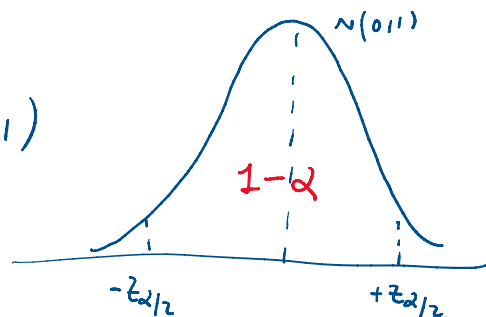


PIVOTAL QUANTITY

POINT ESTIMATION - PARAMETER

$$Z = \frac{(\bar{x} - \bar{y}) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/m_1 + \sigma_2^2/m_2}} \sim N(0,1)$$

COMMON MISTAKE IS TO USE "-" INSTEAD OF "+"



$$P(-z_{\alpha/2} < Z < +z_{\alpha/2}) = 1 - \alpha$$

$$P\left(-z_{\alpha/2} < \frac{(\bar{x} - \bar{y}) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{m_1} + \frac{\sigma_2^2}{m_2}}} < +z_{\alpha/2}\right) = 1 - \alpha$$

... KEEP IN THE MIDDLE $\theta = (\mu_1 - \mu_2)$...

$$P\left((\bar{x} - \bar{y}) - z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{m_1} + \frac{\sigma_2^2}{m_2}} \leq \mu_1 - \mu_2 \leq (\bar{x} - \bar{y}) + z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{m_1} + \frac{\sigma_2^2}{m_2}}\right) = 1 - \alpha$$

LOWER BOUND θ UPPER BOUND

$$\begin{aligned} -z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{m_1} + \frac{\sigma_2^2}{m_2}} &\leq (\bar{x} - \bar{y}) - (\mu_1 - \mu_2) \leq z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{m_1} + \frac{\sigma_2^2}{m_2}} \\ -(\bar{x} - \bar{y}) - z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{m_1} + \frac{\sigma_2^2}{m_2}} &\leq -(\mu_1 - \mu_2) \leq z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{m_1} + \frac{\sigma_2^2}{m_2}} - (\bar{x} - \bar{y}) \\ (\bar{x} - \bar{y}) + z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{m_1} + \frac{\sigma_2^2}{m_2}} &\geq \mu_1 - \mu_2 \geq -z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{m_1} + \frac{\sigma_2^2}{m_2}} + (\bar{x} - \bar{y}) \\ (\bar{x} - \bar{y}) - z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{m_1} + \frac{\sigma_2^2}{m_2}} &\leq \mu_1 - \mu_2 \leq (\bar{x} - \bar{y}) + z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{m_1} + \frac{\sigma_2^2}{m_2}} \end{aligned}$$

$$\Rightarrow C_{1-\alpha}(\mu_1 - \mu_2) = \left[(\bar{x} - \bar{y}) \pm z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{m_1} + \frac{\sigma_2^2}{m_2}} \right]$$

WE NOW REMOVE THE ASSUMPTION THAT σ_1^2 AND σ_2^2 ARE KNOWN

BUT AT LEAST YOU HAVE SOME PARTIAL KNOWLEDGE AND SO YOU KNOW THAT THE TWO ARE EQUAL

2) $\sigma_1^2 = \sigma_2^2$ BUT UNKNOWN (EQUAL UNKNOWN VARIANCES)

THE TWO POPULATION HAVE EQUAL VARIANCES BUT YOU DON'T KNOW WHICH ONE

$\Rightarrow$ WE DO NOT SEPARATE THE TWO SAMPLES X AND Y TO ESTIMATE THE VARIANCES, SINCE ASSUMED EQUAL ($\sigma_1^2 = \sigma_2^2$)

I WILL USE BOTH SAMPLES SO THAT I HAVE MORE DATA TO ESTIMATE THE UNIQUE PARAMETER AS A VARIANCE FOR BOTH POPULATIONS

$\Rightarrow$ POOLED SAMPLE VARIANCE S_C^2 AS UNIQUE ESTIMATOR OF BOTH σ_1^2 AND σ_2^2

$$S_C^2 = \frac{(m_1 - 1)S_1^2 + (m_2 - 1)S_2^2}{m_1 + m_2 - 2}$$

WEIGHTED AVERAGE OF THE TWO SAMPLE VARIANCES

$$S_1^2 = \frac{1}{m_1 - 1} \sum_{i=1}^{m_1} (x_i - \bar{x})^2$$

$$S_2^2 = \frac{1}{m_2 - 1} \sum_{i=1}^{m_2} (y_i - \bar{y})^2$$

MORE IMPORTANCE TO THE LARGER SAMPLE

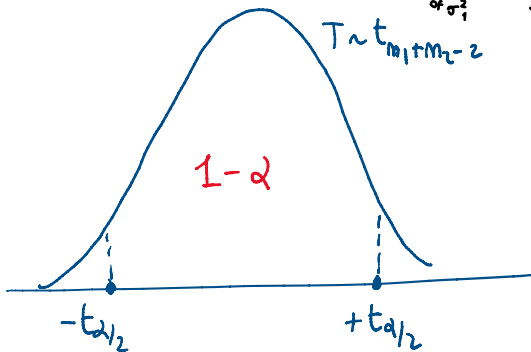
- WEIGHT ASSOCIATED TO S_1^2 IS HIGHER IF m_1 IS HIGHER
- WEIGHT ASSOCIATED TO S_2^2 IS HIGHER IF m_2 IS HIGHER

$$S_c^2 = S_1^2 \cdot \frac{m_1 - 1}{m_1 + m_2 - 2} + S_2^2 \cdot \frac{m_2 - 1}{m_1 + m_2 - 2}$$

$\Rightarrow$ In the CI, S_c^2 replaces both σ_1^2 and σ_2^2 :

$\Rightarrow$ PQ is $T = \frac{(\bar{x} - \bar{y}) - (\mu_1 - \mu_2)}{S_c \sqrt{\frac{1}{m_1} + \frac{1}{m_2}}} \sim t_{m_1 + m_2 - 2}$

Diagram showing the formula with annotations: "TAKE IT OUT OF THE $\sqrt{\quad}$ " pointing to S_c , and "SAMPLE" pointing to S_c^2 . Below the formula, it says "INSTEAD OF σ_1^2 " and "INSTEAD OF σ_2^2 ".



$$P(-t_{\alpha/2} \leq T \leq +t_{\alpha/2}) = 1 - \alpha$$

$$P\left(-t_{\alpha/2} \leq \frac{(\bar{x} - \bar{y}) - (\mu_1 - \mu_2)}{S_c \sqrt{\frac{1}{m_1} + \frac{1}{m_2}}} \leq +t_{\alpha/2}\right) = 1 - \alpha$$

... USUAL STEPS TO KEEP $\mu_1 - \mu_2$

IN THE MIDDLE, TO REACH

THE FORM $P(L_1 \leq \theta \leq L_2) = 1 - \alpha$...

$$CI_{1-\alpha}(\mu_1 - \mu_2) = \left[(\bar{x} - \bar{y}) \mp t_{\alpha/2} S_c \sqrt{\frac{1}{m_1} + \frac{1}{m_2}} \right]$$

QUANTILE OF T-STATIST. WITH $m_1 + m_2 - 2$ DF

(DEPENDENCE ON DF OMITTED FOR BREVITY)

WE ALSO REMOVE THE PARTIAL KNOWLEDGE OF EQUAL VARIANCES

3) σ_1^2 AND σ_2^2 UNKNOWN (POSSIBLY UNEQUAL)

I CAN NOT USE THE POOLED SAMPLE VARIANCE ANYMORE, I NEED TO SEPARATE ESTIMATORS FOR THE TWO SAMPLE VARIANCE

NOT ANYMORE A UNIQUE ESTIMATOR FOR BOTH σ_1^2 AND σ_2^2

$\Rightarrow$ ESTIMATED SEPARATELY BY S_1^2 AND S_2^2

$\Rightarrow$ IN THE PQ: S_1^2 REPLACES σ_1^2

S_2^2 " σ_2^2

$\Rightarrow$ PQ is $T = \frac{(\bar{x} - \bar{y}) - (\mu_1 - \mu_2)}{\sqrt{S_1^2/m_1 + S_2^2/m_2}} \sim t_{\check{\nu}}$, $\check{\nu} < m_1 + m_2 - 2$

Diagram showing a checkmark in a circle next to ν in the denominator, with an arrow pointing to it from the text "DEGREE OF FREEDOMS".

SEE NOTES FOR EXACT VALUE

(DO NOT MEMORIZE)